Question 1

The line $y = x + 1$ intersects the circle $(x-2)^2 + (y-1)^2 = 4$ at two points.
What are the coordinates of the two intersection points? **[1]**

A. $(0,1)$ and $(2,3)$
B. $(0,1)$ and $(4,3)$
C. $(1,2)$ and $(3,4)$
D. $(1,2)$ and $(2,3)$

Question 2

The function f is defined by **[1]**

$$f(x) = x^2 - 6x + 5, 0 \leq x \leq 4$$

The range of f is:
A. $-3 \leq f(x) \leq 5$
B. $-4 \leq f(x) \leq 5$
C. $-4 \leq f(x) \leq 0$
D. $0 \leq f(x) \leq 4$

Question 3

The function f is defined by **[1]**

$$f(x) = 2x^3 - 3x^2 - 12x + 5.$$

Which statement is correct?
A. A local maximum occurs at $x = 2$, and local minimum occurs at $x = -1$.
B. A local maximum occurs at $x = -1$, and local minimum occurs at $x = 2$.
C. A local maximum occurs at $x = 1$, and local minimum occurs at $x = -2$.
D. A local maximum occurs at $x = -2$, and local minimum occurs at $x = 1$.

Question 4

The solution of the inequality $x^2 + x - 6 \geq 0$ is: **[1]**

A. $-3 \leq x \leq 2$
B. $x \leq -3$ or $x \geq 2$
C. $x \leq -2$ or $x \geq 3$
D. $-2 \leq x \leq 3$

Question 5

The line $y = x + k$ does not meet the circle $(x-1)^2 + (y-2)^2 = 9$. **[1]**
The set of possible values of k is:
A. $k < 3 - 3\sqrt{2}$ or $k > 3 + 3\sqrt{2}$
B. $3 - 3\sqrt{2} < k < 3 + 3\sqrt{2}$
C. $k < 1 - 3\sqrt{2}$ or $k > 1 + 3\sqrt{2}$
D. $1 - 3\sqrt{2} < k < 1 + 3\sqrt{2}$

Question 6

(a) Solve the simultaneous equations

$$2p + 3q = 7$$
$$5p - 2q = 8$$

You must show **all** your working. **[3]**

(b) Line l_1 has equation

$$2x + 3y = 7$$

Line l_2 is perpendicular to l_1 and passes through $(4, -1)$.

Find the equation of line l_2 .

Give your answer in the form $ax + by + c = 0$, where a, b, c are integers**.** **[2]**

[Total: 5 marks]

Question 7

(a) Factorise

$$6x^2 - 11x - 10.$$

[2]

(b)

i. Use your answer to *part (a)* to solve

$$6x^2 - 11x - 10 = 0.$$

Give your solutions in the form $\frac{m}{n}$, where m, n are integers. **[1]**

ii. Solve the inequality

$$6x^2 - 11x - 10 > 0$$

[2]

iii. Sketch the graph of

$$y = 6x^2 - 11x - 10.$$

Show clearly the coordinates where the curve crosses the axes. **[2]**

[Total: 7 marks]

Question 8

A circle has equation

$$x^2 + y^2 - 4x + 6y - 3 = 0.$$

(a) Find the radius of the circle and the coordinates of its centre. **[2]**

Point $A\ (2,1)$ and point $B\ (2, -7)$ are the ends of a diameter of the circle.

Point $C\ (2k, -k)$ lies on the circle and is distinct from A and B.

(b)

i. Find expressions, in terms of k, for the gradients of AC and BC. **[2]**

ii. Hence, use the answer to b(i) to find the possible values of k. **[3]**

[Total: 7 marks]

Question 9

The function f is defined by

$$f(x) = x^3 - 3x^2 + 4$$

(a) Use the Remainder Theorem to find the remainder when $f(x)$ is divided by $(x - 1)$

[1]

(b) Divide $f(x)$ by $(x + 1)$ and hence factorise $f(x)$ completely.

[3]

(c) Find the coordinates of the stationary points on the curve $y = f(x)$

[3]

(d) Sketch the graph of $y = f(x)$.
Show clearly the coordinates of all intersections with the coordinates axes and turning points.

[2]

[Total: 9 marks]

Question 10

(a) Given that

$$y = x^3 - 6x^2 - 12x + 5$$

solve the equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$$

[3]

(b) A curve has equation

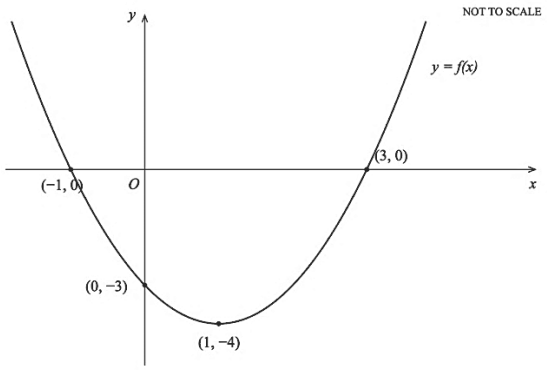
$$\frac{dy}{dx} = \frac{12}{x^2} + 6\sqrt{x} - 1, \qquad x > 0$$

The curve passes through $(4,15)$
Find the equation of the curve.

[3]

[Total: 6 marks]

Question 11



The graph of $y = f(x)$ is an upward-opening parabola with a minimum at $(1, -4)$ and crosses the $x - axis$ at $(-1,0)$ and $(3,0)$.

(a) Sketch the graph of

$$y = f((x - 3) + 1)$$

Show clearly the coordinates of the minimum and both intersections with the $x - axis$

[2]

QUESTIONS CONTINUE ON THE NEXT PAGE

Functions g and h are defined by:

$$g(x) = \frac{4x-5}{3}, x \in R$$
$$h(x) = x^3 + 1, \quad (x \geq 0)$$

(b)
i. Find $g^{-1}(x)$. **[1]**
ii. Evaluate $h'(4)$. **[1]**
iii. Solve

$$g[h(x)] = 85$$

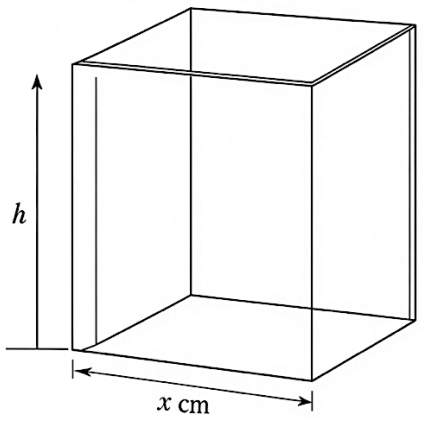
[2]
[Total: 6 marks]

Question 12
The curve C has equation

$$y = x^3 - 12x + 5.$$

(a) Find the coordinates of the stationary points of C. **[3]**
(b) Use the second derivative to determine whether each stationary point is a local maximum or a local minimum. **[2]**
(c) Find the coordinates of the point of inflection of C, justifying that it is a point of inflection. **[2]**
[Total: 7 marks]

Question 13



An open-top box has a square base of side $x\ cm$ and height $h\ cm$.
The total external surface area of the box is $300cm^2$
(a) Show that the volume V of the box, in cm^3, is given by

$$V = 75x - \frac{1}{4}x^3$$

[2]
(b) Find the value of x for which V is stationary.
[2]
(c) Verify that this value of x gives a maximum volume and calculate this maximum volume. **[3]**
[Total: 7 marks]

Question 14

(a) Find

$$\int 4x^3 - \frac{6}{x^2} + 2\sqrt{x}\, dx$$

[3]

(b)

A curve and a straight line have equations

$$y = 4x - x^2$$

and

$$y = x$$

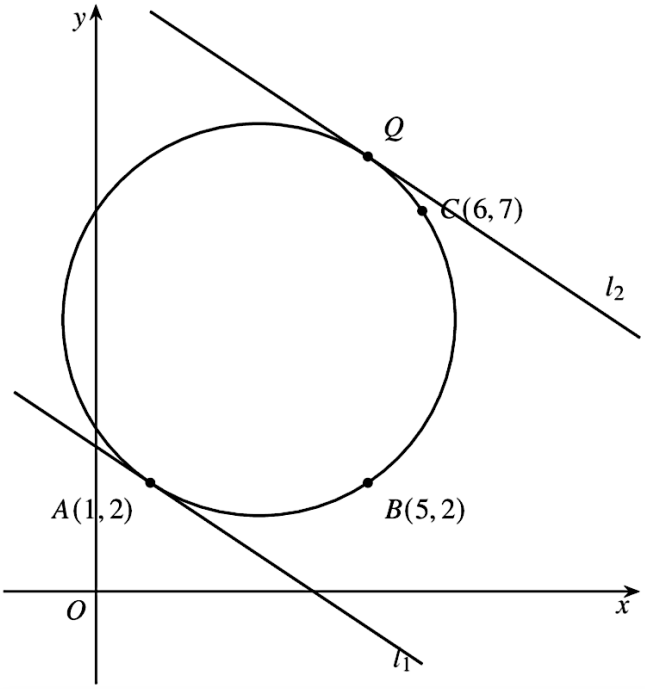
i. Find the coordinates of both intersection points. **[2]**

ii. Find the area of the finite region bounded by the curve and the line. Give your answer to 1 decimal place.

[4]

[Total: 9 marks]

Question 15



Three points $A(1, 2), B(5, 2)$ and $C\ (6, 7)$ lie on a circle.
Line l_1 is tangent to the circle at A.
Line l_2 is the other tangent parallel to l_1 and touches the circle at Q.

(a) Find the coordinates of the centre and the equation of the circle.

[4]

(b) Find the equation of l_1 .

[2]

(c) Find the equation of l_2 and the coordinates of Q.

[5]

(d) Show that AQ is a diameter of the circle.

[1]

[Total: 12 marks]